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Porphyria Cutanea Tarda

Multiplicity of Risk Factors Including *HFE* Mutations, Hepatitis C, and Inherited Uroporphyrinogen Decarboxylase Deficiency

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The coexistence of factors considered to contribute to development of porphyria cutanea tarda was studied in 39 consecutive patients. Highly prevalent factors were alcohol intake in 79%, smoking in 86%, hepatitis C virus infection in 74%, estrogen use in 73% of 11 females, and at least one mutation in the *HFE* (hereditary hemochromatosis) gene in 65%. The C282Y mutation was found in 29%, H63D in 47%, and S65C in 0%. *HFE* genotypes included C282Y/C282Y in 9%, H63D/H63D in 9%, C282Y/H63D in 12%, C282Y/wild type in 9%, and H63D/wild type in 26%. Less prevalent were HIV infection in 15% (or 25% of those tested, $N = 24$) and erythrocyte uroporphyrinogen decarboxylase deficiency, which distinguishes familial (type 2) from "sporadic" (type 1) porphyria cutanea tarda, in 19%. Multiple contributing factors coexisted in both types 1 and 2, with 92% of all patients having three or more factors. These observations indicate that this porphyria is multifactorial in the individual patient, and therefore is seldom attributable to a single identifiable cause. Profiling for all potentially contributing factors is important for individualizing management.

KEY WORDS: porphyria; iron; hemochromatosis; *HFE* gene; hepatitis C; uroporphyrinogen decarboxylase.

Porphyria cutanea tarda (PCT), the most common and readily treated porphyria, is manifested primarily by chronic, blistering lesions over sun-exposed areas of skin. The disease results from the development of marked inhibition or inactivation of liver uroporphyrinogen decarboxylase (UROD, the fifth enzyme in the heme biosynthetic pathway), by an iron-dependent process, and oxidation of the substrate uroporphyrinogen (a reduced porphyrin) to uroporphyrin. Porphyrins, especially uroporphyrin (octacarboxyl porphyrin) and heptacarboxyl porphyrin accumulate in liver, are transported in plasma to other tissues, including skin, and excreted in urine and feces. Photoactivation of porphyrins by long-wave ultraviolet light leads to skin damage (1, 2).

Iron plays a key role in PCT, as indicated by certain clinical features and experimental studies. The amount of UROD protein measured immunochemically in the liver is not decreased, suggesting an acquired process that inactivates or inhibits the enzyme (3). Repeated phlebotomy to reduce hepatic iron leads to clinical remission and at least partially restores hepatic UROD activity (3, 4). Uroporphyria in rodents is enhanced by iron overload and prevented by iron deficiency. The effect of iron is probably not

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mediated by direct inhibition of UROD, but rather by generation of reactive oxygen species that can oxidize uroporphyrinogen to uroporphyrin and to uncharacterized, nonporphyrin products capable of inhibiting UROD (2, 5, 6).

Multiple factors are thought to contribute to the pathogenesis of PCT (2). This is plausible given that this uncommon condition develops in very few individuals with each of the individual environmental and genetic factors that are thought to contribute. Because there is little information regarding the number of factors that may be expected to coexist in individual patients with PCT, it was of interest to collect nearly complete information on multiple putative risk factors in a consecutive series of 39 PCT cases. All patients were seen at a single academic medical center where the diagnosis was established in all cases by testing in a specialty laboratory using published methods. This study demonstrates multiplicity of risk factors in PCT and that usually three or more are expected to coexist in the great majority of individual patients.

MATERIALS AND METHODS

Thirty-nine consecutive patients with PCT were seen either initially or at follow-up in the Gastroenterology Clinic or the General Clinical Research Center of the University of Texas Medical Branch (UTMB) at Galveston within a 1.5-year period. The UTMB Institutional Review Board approved the study, and informed consent was obtained. All patients but one patient were from the surrounding region of eastern Texas rather than referrals from more distant locations. Patients were asked whether they smoked, drank alcohol regularly (at least an average intake of 10 g alcohol/day for one year), or used estrogens. Drugs other than estrogens are seldom implicated in contributing to PCT (1), and none of our patients was taking drugs implicated in contributing to acute hepatic porphyrias. PCT was documented in the Porphyria Laboratory at UTMB by substantial increases in urinary uroporphyrin (octacarboxyl porphyrin) and heptacarboxyl porphyrin, increased plasma porphyrins (with a characteristic fluorescence emission spectrum at neutral pH) (7, 8), and/or increased fecal content of isocoproporphyrins.

Total plasma porphyrin concentration was measured by spectrofluorometry (9) and the neutral fluorescence emission spectrum of plasma diluted with phosphate buffer at neutral pH was recorded as described by Poh-Fitzpatrick (7, 8). Urinary δ-aminolevulinic acid and porphobilinogen (10) and total urinary porphyrins were measured as previously described (11) except that porphyrins were determined by spectrofluorometry. Fecal porphyrins were determined as described by Lockwood et al (12) after extraction with ethyl ether, reextraction with HCl, dilution with water, and measurement by fluorescence spectrophotometry. Porphyrins in plasma, urine, and feces were separated by reverse-phase HPLC and fluorescence detection (13). Erythrocyte UROD activity was measured using whole blood lysates as described by McManus et al (14) using uroporphyrinogen I as substrate.

DNA was extracted from blood samples in the UTMB Molecular Pathology Laboratory and used for detecting the C282Y, H63D, and S65C mutations of the *HFE* gene. Polymerase chain reaction (PCR) was performed using two sets of primers directed to the C282Y and H63D alleles. The fragments were digested with restriction enzymes (*RsaI*/*SnaBI* for C282Y and Bc*IMboI* for H63D) and the resulting restriction fragment length polymorphism (RFLP) patterns were separated by agarose gel electrophoresis and stained by ethidium bromide. The restriction enzyme HinfI was used to identify the S65C mutation in the same PCR product that was used to identify the H63D mutation. Standard blood tests and serologies, including anti-HCV and anti-HIV assays, were performed by the UTMB Clinical Laboratory.

Data are presented as mean ± SD and as median (range). The chi-square test was used to assess differences in proportions.

RESULTS

The frequencies of risk factors that we studied are summarized in Table 1. These include environmental factors (alcohol, smoking and estrogens), chronic viral infections (HCV and HIV), and genetic factors (C282Y, H63D, and S65C mutations of the *HFE* gene, and inherited deficiency of UROD). Alcohol, smoking, HCV infection, and the C282Y and H63D *HFE* mutations were highly prevalent in this case series, whereas estrogen use, HIV, and erythrocyte UROD deficiency were less common (Table 1). Estrogen intake was noted in only 21% of patients. However, all were female, and the frequency of a history of estrogen usage in female patients with PCT was high (73%). None had the S65C mutation of the *HFE* gene.

In profiling individual patients for coexisting risk factors, which was the main focus of this study, patients with the C282Y/C282Y, H63D/H63D, or C282Y/H63D genotypes were considered to have two risk factors, and patients who were heterozygous for either mutation to have one risk factor. Both the heterozygous and homozygous states for *HFE* mutations appear to be risk factors for PCT (see later discussion), and it may be argued that in either case the risk factor may be excess iron. We were unable to assess individual degrees of iron overload in our patients before treatment because phlebotomies were usually initiated or completed elsewhere, initial and final ferritin and transferrin saturation values were not obtained, and few patients underwent liver biopsy before treatment of PCT. However, it is also well

TABLE 1. FREQUENCIES OF DISEASE-ASSOCIATED RISK FACTORS IN A SERIES OF 39 PATIENTS WITH PORPHYRIA CUTANEA TARDA

Risk factor	*Patients tested (*N)	*Risk factor present [*N *(%)]*		
		All patients	*HCV-positive patients*	*HCV-negative patients*
Environmental				
Alcohol use	38	30 (79%)	26 (93%)*	4 (40%)
Smoking	35	30 (86%)	25 (96%)*	5 (56%)
Estrogen use	39	8 (21%)	5 (17%)	3 (30%)
Females	11	8 (73%)	5 (100%)	3 (50%)
Infectious				
Hepatitis C	39	29 (74%)		
HIV	24	6 (25%)	2 (12%)*	4 (57%)*
Genetic				
HFE mutations†	34	22 (65%)	16 (62%)	6 (75%)
C282Y‡	34	10 (29%)	8 (31%)	2 (25%)
H63D‡	34	16 (47%)	11 (42%)	5 (63%)
S65C‡	34	0	0	0
Low erythrocyte UROD	36	7 (19%)	4 (15%)	1 (10%)

*Significant difference between HCV-positive and HCV-negative patients.
†Patients with at least one *HFE* mutations.
‡Patients with at least one C282Y, H63D, or S65C allele.

established that inheriting two *HFE* mutations (one from each parent) contributes considerably more to iron overload than does inheriting one mutation (from only one parent).

Multiple contributing factors coexisted in almost all patients (Figure 1). Only three patients (8%) had as few as one or two coexisting risk factors. (Information on all putative risk factors was available in two of these three patients, one had information missing on two factors.) Most patients (72%) had three (17 of 39 patients) or four (11 of 39 patients) risk factors. Eight patients (21%) had as many as five to seven risk factors (Figure 1).

Most patients in this series were Caucasian (95%) and male (72%) and were commonly (74%) infected with HCV. Average age of onset of symptoms for the group of 39 patients was 41.5 ± 8.6 years (median 42 years, range 22–73 years), duration of symptoms prior to evaluation averaged 2.7 ± 4.6 years (median 1.0 years, range 0.1–17 years), and these were not different in those with and without HCV infection (not shown). HCV infection was more common in male [24 of 29, (83%)] than in female patients with PCT [4 of 10 (40%), $P = 0.0096$, chi-square test]. Most patients smoked cigarettes (86%) and used alcohol (79%). Smoking and alcohol use were more prevalent in the HCV-positive group ($P = 0.0027$ and 0.0004, respectively by chi-square test). *HFE* mutations (C282Y and/or H63D) were present in 22 of 34 patients (65%) and, as discussed below, differences in frequencies of these mutations in patients with and without HCV infection were not statistically significant. HIV infection was present in 6 (25%) of the 24 patients tested, and most [4 (67%)] of those infected with HIV were HCV negative ($P = 0.04$). In all but one patient, HIV infection was recognized prior to the onset and diagnosis of PCT. If it is assumed, as seems likely, that of the 15 patients not tested for HIV few or none were HIV-infected, the overall prevalence of HIV infection in this series of PCT patients would be not less, and probably not much greater, than 15%.

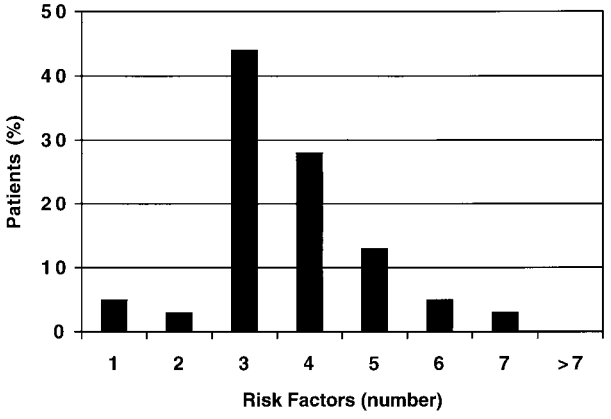


Fig 1. Numbers of coexisting factors considered to contribute to development of porphyria cutanea tarda in a series of 39 patients.

Frequencies of *HFE* genotypes in 34 patients are shown in Table 2. Although the frequencies of the C282Y and H63D mutations and genotypes were mostly higher in the HCV-negative patients, these differences were not statistically significant (Tables 1 and 2, chi-square test or Fisher's exact test, as appro-

Table 2. Frequencies of *HFE* Genotypes in 34 Patients with Porphyria Cutanea Tarda, 26 of Whom Were Infected with Hepatitis C Virus

		*Frequency [*N *(%)]*	
Genotype	*All patients (*N = *34)*	*HCV-positive patients (*N = *26)*	*HCV-negative patients (*N = *8)*
C282Y/C282Y	3 (9%)	2 (8%)	1 (13%)
C282Y/wild type	3 (9%)	3 (12%)	0 (0%)
C282Y/H63D	4 (12%)	3 (12%)	1 (13%)
H63D/wild type	9 (26%)	7 (27%)	2 (25%)
H63D/H63D	3 (9%)	1 (4%)	2 (25%)
Wild type/wild type	12 (35%)	10 (38%)	2 (25%)
Totals	34 (100%)	26 (100%)	8 (100%)

priate). For example, 6 of the 8 (75%) HCV-negative patients had at least one *HFE* mutation compared to 16 of 26 (62%) of HCV-positive patients ($P = 0.49$, chi-square test). The H63D/H63D genotype seemed more prevalent in the HCV-negative patients, and this difference approached statistical significance ($P = 0.0595$).

Partial erythrocyte UROD deficiency (<33 nmol/ml/RBC/hr), which reflects an inherited deficiency of this enzyme, was found in 7 of 36 patients (19%, 3 men, 4 women, Table 1). These patients were classified as having familial (type 2) PCT. Additional risk factors were present in all of these patients. Two patients had two, and the other five patients each had three to six factors in addition to partial erythrocyte UROD deficiency. *HFE* mutations were present in five of seven (71%, homozygous C282Y in one, homozygous H63D in two, and heterozygous H63D in two patients), HCV in five, estrogen use in three, ethanol use in four, and smoking in five.

DISCUSSION

The prevalence and, more importantly, the multiplicity or coexistence of putative risk factors were assessed in 39 patients with PCT seen at a single center. Alcohol intake, smoking, HCV infection, and *HFE* mutations occurred in 65–86%. Estrogen use was highly prevalent in female patients, but was not observed in males. Estrogen-associated PCT in males was reported in the past, when estrogen treatment for advanced prostate cancer was common (1, 15). Infection with HIV and an inherited deficiency of UROD were found less commonly. As discussed further below, alcohol, estrogen, and homozygous *HFE* mutations are accepted as contributing to human PCT, whereas smoking, heterozygous *HFE* mutations, and HIV are less definitely established. The aims of this study did not include establishing the relative importance of these factors in causing PCT, determining their mechanisms, or the strength of interactions among them, rather the aim was to determine how frequently they coexist in individual patients. Exclusion of some factors from the analysis, or considering the inheritance of an *HFE* mutation from each parent as representing one risk factor rather than two, would not alter the main finding of this study, namely that multiple contributing factors in individual patients is a characteristic feature of PCT.

More than one of these factors was found in the great majority (92%) of PCT patients (Figure 1). Most had three or four risk factors [28, or (72%)], and 21% had five to seven. Therefore, because PCT is multifactorial even in the individual patient, a single identified factor should not be considered the sole contributing cause, and multiple contributing factors should always be sought. The number of risk factors in individual patients is likely to be even greater than indicated in this study. Risk factor data were incomplete in a few patients, which would underestimate somewhat the number of factors identified. Other contributors, such as deficiencies of antioxidants (eg, such as vitamin C and carotenoids) (16, 17), were not investigated in this study, and there are likely to be others that are not yet known.

Multiple infectious and environmental factors may contribute to PCT by influencing oxidative stress in hepatocytes. For example, hepatic iron may be increased with HCV infection, and this may enhance oxidative stress (18). HCV and chlorinated aromatic hydrocarbons may release storage iron in hepatocytes in a form that enhances formation of reactive oxygen species (2, 19). Alcohol and estrogens generate active oxygen in some experimental systems (20, 21). Uroporphyria induced by hexachlorobenzene in rodents is accentuated by estrogens and attenuated by vitamin C (22, 23). This antioxidant prevents the CYP1A2-mediated oxidation of uroporphyrinogen to uroporphyrin, the accumulation of uroporphyrin in hepato-

cyte cultures (24), and the development of experimental uroporphyria in a vitamin C-requiring strain of rats (25). Low circulating levels of vitamin C and carotenoids are reported in some PCT patients (16, 17).

Smoking may contribute by inducing cytochrome P-450 enzymes, which can generate reactive oxygen species that damage UROD or oxidize uroporphyrinogen to uroporphyrin. Experimental uroporphyria induced or enhanced in rodents and hepatocyte cultures by certain chemicals (chlorinated cyclic hydrocarbons and polycyclic aromatic hydrocarbons) is substantially influenced by these enzymes. These chemicals also induce the CYP1A subfamily of the cytochrome P-450 enzymes (26). CYP1A2 in rodents catalyzes the oxidation of uroporphyrinogen to uroporphyrin, and this process is enhanced by iron (27–29). CYP1A2 knockout mice are highly resistant to developing uroporphyria (30). Although recent studies suggest that most uroporphyrinogen oxidase activity in human liver is not attributable to CYP1A2, other P-450 enzymes, including the CYP2E subfamily, which are induced by alcohol, may be involved (31).

Although prevalences of HCV in PCT vary considerably by geographic location (32–35), high prevalences in many locations strongly suggest a causal association. The prevalence of HCV infection in this series [29 of 39 patients, (74%)] is similar to reports from southern Europe and greater than reported in two other large studies in North America (56% and 59%) (36, 37). In a recent smaller series of 26 patients in Indiana, 94% of 17 patients tested were HCV-infected (35). The lower prevalences of this infection in PCT in northern Europe (<20%) (38, 39) suggest that in such areas PCT and HCV may be less common or that risk factors other than HCV, such as excess alcohol use, may be more prevalent in the general population and in PCT. HCV is common in the general patient population seen at this medical center. HCV-positive patients in this series were most commonly males, but their age and duration of symptoms did not differ significantly from HCV-negative patients. Most PCT patients used alcohol (79%) and smoked cigarettes (86%), and these habits were even more prevalent among those infected with HCV (93% and 96%, respectively) than in HCV-negative patients (40% and 56%, respectively, Table 1).

HIV infection was noted in 6 of 24 (25%) PCT patients tested, or not less than 15% overall (Table 1). Four of the 6 HIV-positive patients (67%) had two or more additional risk factors, although only 2 (33%) were coinfected with HCV. There have been many previous reports suggesting an association between HIV infection and PCT (40–43), although the mechanisms whereby HIV itself might lead to UROD deficiency in the liver are not clear. Because other risk factors are common in PCT patients with HIV, as confirmed in this study, it has been suggested that HIV infection is not a true contributing factor (42). However, this conclusion does not seem warranted because, as shown in this study, PCT is multifactorial in the presence or absence of HIV, and the presence of one risk factor (such as HCV) should not lead to the conclusion that another associated factor (such as HIV) has no influence. It is plausible, therefore, that HIV is one of the multiple causative factors in this disease, even if less commonly present than several others.

The C282Y and H63D mutations of the hemochromatosis *HFE* gene are common in many populations and have been associated with hereditary hemochromatosis. These mutations are in complete linkage disequilibrium, and therefore never occur on the same chromosome (44). The C282Y mutation is especially prevalent in populations of northwestern European origin (10–20% carry at least one copy) and much less so in southern and eastern Europe (2–4% carry at least one copy). The H63D mutation is more common than the C282Y mutation throughout Europe, with a prevalence of 15–40% (45). Based on studies in northern European and North American populations, allelic frequencies for C282Y and H63D are estimated to be 6.8% and 17.6%, respectively (45). The C282Y/C282Y genotype was identified in 60–90% (average ≈85%) of white hemochromatosis patients (45). Penetrance of the C282Y mutation is stronger than that of the H63D mutation. Individuals with the C282Y/H63D and H63D/H63D genotypes are at risk for iron overload, but apparently less so than for the C282Y/C282Y genotype (44, 46–48). The uncommon and much weaker S65C mutation only occasionally contributes to iron overload (49–54). The association of the C282Y and H63D mutations with PCT (both types 1 and 2) is one indication that liver iron content is an important influence in this disease (36, 37, 55–60). Some investigators have reported a higher than expected prevalence of the C282Y mutation in PCT (37, 55, 58) and others increased frequencies of both the C282Y and H63D mutations (36, 59). The S65C mutation has, to our knowledge, been reported in only one PCT patient—a C282Y/S65C compound heterozygote (50).

Frequencies of the *HFE* genotypes and allele fre-

TABLE 3. COMPARISONS OF *HFE* GENE MUTATIONS IN PATIENTS WITH PORPHYRIA CUTANEA TARDA

		HFE *genotype [*N *(%)]*					*Allele frequency (%)†*	
Country (ref)	N	*C282Y/C282Y*	*C282Y/WT**	*H63D/H63D*	*H63D/WT**	*C282Y/H63D*	*C282Y*	*H63D*
USA								
This study	34	3 (9)	3 (9)	3 (9)	9 (26)	4 (12)	27	35
Bonkovsky (36)	26	5 (19)	6 (23)	2 (8)	6 (23)	0 (0)	38	28
Bulaj (37)	87	17 (20)	13 (15)	6 (7)	13 (15)	6 (7)	37	26
UK(55)	41	7 (17)	8 (20)	1 (2)	9 (22)	3 (7)	37	25
Australia(58)	27	3 (11)	4 (15)	2 (7)	5 (19)	5 (19)	38	36
Italy(60)	68	0 (0)	1 (1.5)	5 (7)	28 (41)	1 (1.5)	2.1	29
Spain(61)	88	3 (3)	5 (6)	10 (11)	24 (27)	7 (8)	14	32
Bulgaria(65)	48	0 (0)	0 (0)	0 (0)	10 (21)	1 (2)	1.2	12

*WT; wild type.

†The C383Y and H63D mutations are in complete linkage disequilibrium. Therefore, when allele frequency was calculated, chromosomes bearing the other mutant allele were not considered "at risk."

quencies in this and previous PCT series are compared in Table 3. The occurrence of at least one *HFE* mutation in 22 of 34 patients (65%) in this series is similar to the finding of *HFE* mutations in 19 of 26 PCT patients (73%) and 55 of 87 patients (63%) in two previous US series (36, 37). The C282Y mutation was more common in this case series (allele frequency 27%) than was observed in Italy and Spain (allele frequencies 2% and 14%, respectively), and somewhat less common than reported in the two previous US, UK, and Australia series (allele frequencies 37–38%, Table 3) (36, 37, 58, 60, 61). These C282Y allelic frequencies in PCT in the US, UK, and Australia series are all higher than the expected population frequency of approximately 6.8% (45). On the other hand, the H63D allele frequency of 35% in this series of PCT patients was similar to those reported from Italy, Spain, Australia, the UK, and two previous US series (25–36%) (36, 58, 60, 61). Except for the UK and previous US studies, these H63D allelic frequencies are all higher than an expected frequency of about 17.6% (45). The *HFE* mutation frequencies in this study may reflect the largely Northern European ancestry common among Caucasian residents of this area of Texas, as well as elsewhere in the United States, and in Australia. No patients in this series were black and only two were Hispanic (neither had *HFE* mutations), and it is not known whether prevalences of *HFE* mutations are different among PCT patients from these ethnic groups in the United States. *HFE* mutations were somewhat more frequent in the HCV-negative patients (75%) than in HCV-positive patients (62%), but the differences were not statistically significant (Tables 1 and 2). Recent studies suggest that iron accumulation, although common in chronic hepatitis C, is usually not related to *HFE* mutations (62, 63).

The prevalence of low erythrocyte UROD in this series (7 of 36 patients, 19%, Table 1) was similar to other PCT series (1, 2). Multiplicity of risk factors was a feature of patients with this trait, who were classified as having familial (type 2) PCT (2). Five of seven patients with this trait had three to six additional risk factors (see Results). Half-normal erythrocyte UROD activity identifies individuals with heterozygous mutations of the UROD gene (2, 36), who from birth have reduced amounts of enzyme protein in all tissues and are more susceptible to develop the profound degree of hepatic UROD deficiency that is needed to cause clinically manifest PCT. In sporadic (type 1) PCT, erythrocyte UROD is normal, and no mutations of the UROD gene have been identified (64). A rare type 3 PCT, which was not identified in this study, resembles type 1 biochemically but more than one family member is affected (1, 2). Major clinical features and treatment of these types of PCT are identical. This study indicates that the coexistence of multiple risk factors represents an additional common feature of types 1 and 2 PCT.

An important implication of this study is that multiple risk factors should be sought in all patients with PCT, including those with familial (type 2) PCT. The particular factors identified by profiling each PCT patient for known or putative risk factors may influence treatment. For example, the finding of coexisting HCV or HIV will likely lead to specific antiviral treatment. Although repeated phlebotomy and low-dose chloroquine are both standard therapies for PCT, the former is preferred, especially in patients with marked iron overload and *HFE* genotypes such

as C282Y/C282Y or C282Y/H63D. Cessation of alcohol and possibly smoking can be reasonably considered to reduce the chances of disease recurrence after standard therapy. In women, estrogen therapy is usually stopped at least until remission of PCT is achieved by standard therapy.

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